Molecular Graph Learning under Different Scientific Data Conditions: A Study of Chemprop MPNN for Aqueous Solubility and Lipophilicity Prediction

2351458 Ren Hongzhen 2352979 Yu Shu 2353117 Huang Yanwei
2353814 Ma Xiaolong

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Abstract

Predicting absorption, distribution, metabolism, excretion and toxicity (ADMET) related properties directly from molecular structure is a representative *AI for Science* problem in early drug discovery. In this report we study how a message passing neural network (Chemprop MPNN) behaves under different scientific data conditions: dataset size, molecular property type, modelling paradigm, and the way the train/test split is constructed. We evaluate two MoleculeNet benchmark datasets — ESOL aqueous solubility (1,128 molecules, target $\log S$) and Lipophilicity (4,200 molecules, target $\log D$) — against a traditional RDKit-descriptor Random Forest baseline, under both random and Bemis–Murcko scaffold splits. On a fixed held-out test set the Chemprop MPNN reaches RMSE $= 0.612$, $R^2 = 0.921$ on ESOL and RMSE $= 0.550$, $R^2 = 0.796$ on Lipophilicity. Performance improves monotonically with training-set size on both tasks, the MPNN outperforms the Random Forest baseline under random splits, and we observe a clear and *property-dependent* effect of scaffold splitting: ESOL degrades modestly while Lipophilicity is largely robust. We discuss what these patterns imply about the reliability of scientific machine-learning models when chemical coverage is limited.

1. Introduction

A central promise of *AI for Science* is to replace slow, expensive physical measurement with fast, structure-based prediction. In drug discovery this matters acutely: a candidate molecule must not only bind its target but also possess acceptable ADMET properties. Two of the most fundamental such properties are **aqueous solubility** and **lipophilicity**. Solubility governs whether a compound can dissolve and be absorbed at all; lipophilicity (commonly summarised by the distribution coefficient $\log D$) governs membrane permeability and non-specific binding. The two properties are coupled by a well known trade-off: too little lipophilicity impairs permeability, while too much harms solubility and selectivity.

Classical solubility models such as the ESOL equation of Delaney [1] estimate $\log S$ from a small set of hand-chosen descriptors. Modern graph neural networks instead *learn* the molecular representation directly from the molecular graph. The Chemprop directed message passing neural network of Yang et al. [2], and its modular successors [3,4], are the de-facto open-source standard for this approach.

A single accuracy number on a single dataset, however, says little about whether such a model is *scientifically reliable*. This report therefore asks one consolidated research question:

> **How does molecular graph learning perform under different scientific data conditions — namely data scale, molecular property type, modelling paradigm, and chemical generalisation split?**

We address this through four coordinated experiments on two datasets, comparing a learned graph model against a transparent descriptor baseline, and contrasting random splits against scaffold splits that emulate the real discovery setting of encountering novel chemotypes.

2. Datasets

We use two public MoleculeNet [5] regression benchmarks. Both are stored in a unified two-column format (`smiles`, `target`) so that the same training pipeline applies without hard-coding the target name.

ESOL (Delaney). 1,128 organic small molecules with experimentally measured aqueous solubility $\log S$ ($\log_{10}$ mol/L). Source: Delaney 2004 [1]. The target ranges from -11.60 to 1.58 with mean -3.05 and standard deviation 2.10.

Lipophilicity. 4,200 drug-like molecules with experimental octanol/water distribution coefficient $\log D$ at pH 7.4, taken from the DeepChem/MoleculeNet Lipophilicity CSV. The target ranges from -1.50 to 4.50 with mean 2.19 and standard deviation 1.20.

Table 1: Summary of the two benchmark datasets.

Dataset	Task	Molecules	Target	Range	Metrics
ESOL	Aqueous solubility	1,128	$\log S$	$[-11.60, 1.58]$	RMSE, MAE, R^2
Lipophilicity	Lipophilicity / $\log D$	4,200	$\log D$	$[-1.50, 4.50]$	RMSE, MAE, R^2

Splitting strategies. Each dataset is split 80/20 into train/test. We construct two kinds of split:

- **Random split.** Molecules are assigned to train/test uniformly at random. This tests *interpolation* among chemically similar molecules.
- **Scaffold split.** Molecules are grouped by their Bemis–Murcko scaffold [6] (RDKit `MurckoScaffold.MurckoScaffoldSmiles`); whole scaffold groups are assigned to either train or test, largest groups first. This tests *generalisation* to chemical skeletons that were never seen during training, which is the situation actually faced when screening novel series in drug discovery.

After splitting, ESOL yields 902 train / 226 test molecules and Lipophilicity yields 3,360 train / 840 test molecules. During training the train portion is further split 90/10 into an internal training/validation set (e.g. ESOL: 811 train / 91 validation).

3. Methods

3.1 Chemprop MPNN

The primary model is a directed-bond message passing neural network as implemented in Chemprop v2 [3,4]. Each molecule is represented as a graph whose atoms and bonds carry feature vectors. The model performs T rounds of bond-centred message passing; the hidden state of bond $(u \to v)$ at step $t+1$ is

$$h_{u\to v}^{(t+1)} \;=\; \tau\Big(h_{u\to v}^{(0)} \;+\; W \sum_{w\in\mathcal{N}(u)\setminus v} h_{w\to u}^{(t)}\Big),$$

where $\mathcal{N}(u)$ is the set of neighbours of atom u, W a learned weight matrix and τ a non-linearity. After T steps the bond states are aggregated into atom states and then mean-pooled into a single molecular vector h_G, which a feed-forward network maps to the scalar property. The network is trained end-to-end by minimising the mean squared error

$$\mathcal{L} \;=\; \frac{1}{N}\sum_{i=1}^{N}\big(\hat{y}_i - y_i\big)^2.$$

Hyper-parameters are held fixed across all datasets and splits to keep comparisons clean: hidden size 300, message-passing depth 5, dropout 0.1, batch normalisation enabled, batch size 64, Adam optimiser with a plateau learning-rate scheduler, maximum 100 epochs, and early stopping with patience 15 on the validation loss.

3.2 Random Forest baseline

To contrast learned representations against classical feature engineering, we use a Random Forest regressor [7] on eight interpretable RDKit [8] descriptors: molecular weight (`MolWt`), Crippen log P (`MolLogP`), topological polar surface area (`TPSA`), hydrogen-bond donors (`NumHDonors`) and acceptors (`NumHAcceptors`), rotatable bonds (`NumRotatableBonds`), ring count (`RingCount`) and the fraction of sp^3 carbons (`FractionCSP3`). The baseline is evaluated on exactly the same train/test molecules as Chemprop.

3.3 Evaluation metrics

For N test molecules with true values y_i, predictions $\hat{y}_i$ and mean $\bar{y}$, we report

$$\mathrm{RMSE} = \sqrt{\tfrac{1}{N}\textstyle\sum_i(\hat{y}_i - y_i)^2}, \quad \mathrm{MAE} = \tfrac{1}{N}\textstyle\sum_i|\hat{y}_i - y_i|, \quad R^2 = 1 - \frac{\sum_i(\hat{y}_i-y_i)^2}{\sum_i(y_i-\bar{y})^2}.$$

3.4 Experimental design

We run four experiments: (E1) Chemprop baseline on both datasets under the random split; (E2) a data-scale experiment training on 20 %, 50 %, 80 % and 100 % of the training pool, three repetitions each with different seeds, on a fixed test set; (E3) Chemprop vs. Random Forest; and (E4) random split vs. scaffold split for both models. A qualitative case study on common drug molecules complements the quantitative results.

4. Experiments and Results

4.1 E1 — Chemprop baseline on both properties

Table 2 reports the held-out test performance of the Chemprop MPNN under the random split. The model explains 92 % of the variance in ESOL solubility and 80 % of the variance in Lipophilicity $\log D$.

Table 2: Chemprop MPNN baseline (random split, fixed test set).

Dataset	n_{train}	n_{test}	RMSE	MAE	R^2
ESOL	811	226	0.612	0.452	0.921
Lipophilicity	3,024	840	0.550	0.405	0.796

Figures 1 and 2 show predicted-versus-measured density plots. For ESOL the fitted slope is 0.95, very close to the ideal diagonal, indicating almost no systematic bias across four orders of magnitude in solubility. For Lipophilicity the slope is 0.81, showing mild *regression to the mean* (high $\log D$ slightly under-predicted, low $\log D$ slightly over-predicted), consistent with the narrower dynamic range and the additional physical complexity of $\log D$ (it depends on the ionisation state of the molecule at a given pH, not only on its neutral structure).

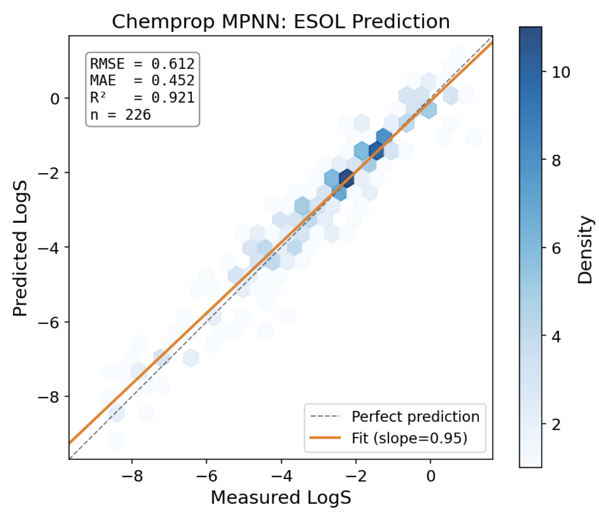

Figure 1: Chemprop MPNN on ESOL: predicted vs. measured $\log S$ (random split, n=226).

The two tasks are not equally hard. ESOL achieves a higher R^2 despite a *larger* RMSE: this is expected because solubility spans a much wider value range (std 2.10) than $\log D$ (std 1.20), so RMSE alone is not comparable across tasks — R^2 is the fairer cross-task indicator.

4.2 E1b — Error structure of the baseline model

A single RMSE figure hides *how* a model is wrong. To characterise the error structure we examined the distribution of the signed residual $e_i = \hat{y}_i - y_i$ on the ESOL test set (Figure 3). The residuals are approximately Gaussian, centred almost exactly at zero (mean error = 0.080, standard deviation = 0.607, MAE = 0.452). The near-zero mean indicates the model is essentially *unbiased*: it does not systematically over- or under-predict solubility, consistent with the fitted slope of 0.95 in Figure

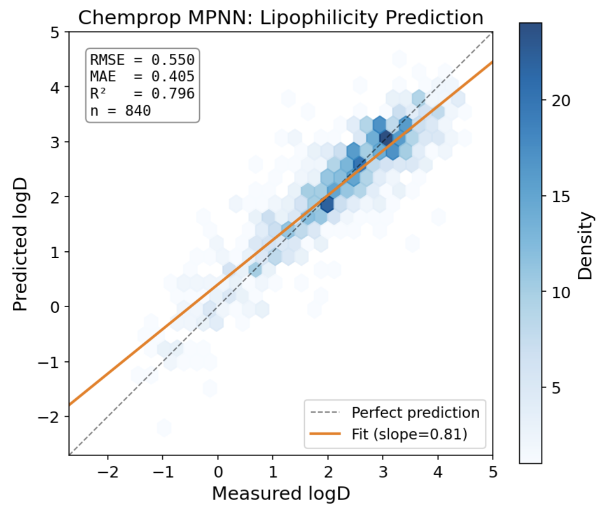

Figure 2: Chemprop MPNN on Lipophilicity: predicted vs. measured $\log D$ (random split, n=840).

1. Quantitatively, 67.3 % of test molecules are predicted within ± 0.5 log units and 88.1 % within ± 1.0 log unit of the measured value.

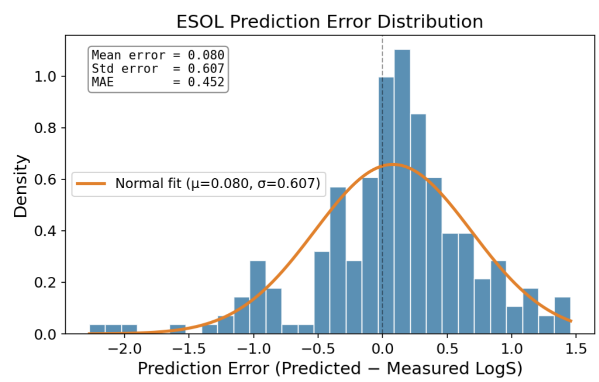

Figure 3: ESOL prediction-error distribution on the test set (n=226); a near-Gaussian, zero-centred residual with $\sigma \approx 0.61$.

Two observations matter for scientific interpretation. First, the residual spread ($\sigma \approx 0.61$ log units) is comparable to the experimental reproducibility commonly reported for aqueous-solubility measurements (often cited at the level of a few tenths of a log unit [1]). The model error is therefore approaching the *intrinsic noise floor* of the labels rather than reflecting only model deficiency — further accuracy gains would increasingly require cleaner data, not merely a larger network. Second, the residual tail is mildly asymmetric: the largest under-prediction reaches -2.27 log units whereas the largest over-prediction is only $+1.46$. The heavier negative tail corresponds to a few very poorly soluble compounds whose extreme $\log S$ values are under-represented in training, an expected small-data coverage effect that re-appears in §4.5.

4.3 E2 — Effect of training-data scale

We trained Chemprop on 20 %, 50 %, 80 % and 100 % of the available training molecules (three seeds each) against a fixed test set. Table 3 and Figures 4–5 summarise the mean metrics; error bars are the standard deviation over the three repetitions.

Table 3: Data-scale experiment. Values are mean over 3 seeds (RMSE std in parentheses).

Dataset	Scale	n_{train}	RMSE (std)	R^2
ESOL	20 %	162	0.977 (0.059)	0.797
ESOL	50 %	405	0.731 (0.012)	0.887
ESOL	80 %	648	0.675 (0.044)	0.903
ESOL	100 %	811	0.643 (0.031)	0.912
Lipophilicity	20 %	604	0.790 (0.029)	0.577
Lipophilicity	50 %	1,512	0.640 (0.013)	0.723
Lipophilicity	80 %	2,419	0.588 (0.017)	0.766
Lipophilicity	100 %	3,024	0.558 (0.003)	0.789

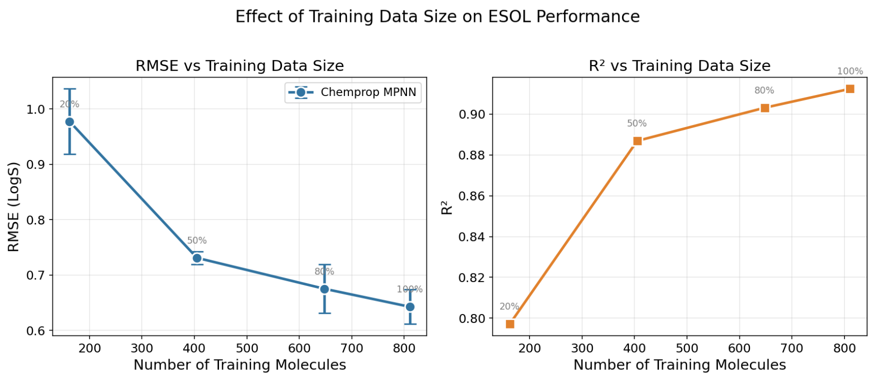

Figure 4: Effect of training-set size on ESOL performance (mean ± std over 3 seeds).

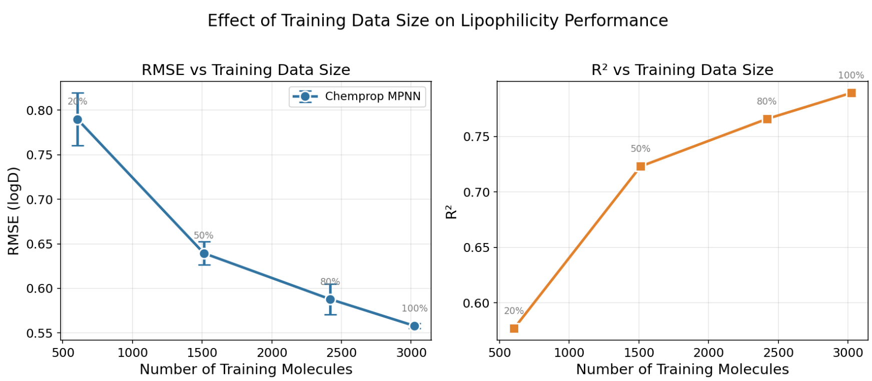

Figure 5: Effect of training-set size on Lipophilicity performance (mean ± std over 3 seeds).

On both tasks, RMSE decreases and R^2 increase monotonically with training size, but with **diminishing returns**: the ESOL RMSE drops sharply from 20 % to 50 % ($0.977 \to 0.731$) and then flattens ($0.675 \to 0.643$ from 80 % to 100 %); Lipophilicity shows the same saturating shape. The variance across seeds also shrinks as more data is added — the Lipophilicity RMSE std falls from 0.029 at 20 % to 0.003 at 100 % — so larger datasets yield not only more accurate but also more *stable* models. This is the central empirical message of the project: in molecular property prediction, data quantity is a first-order determinant of both accuracy and reproducibility.

A useful corollary is the *efficiency* of each additional molecule. Going from 20 % to 50 % of the ESOL training pool (an extra 243 molecules) lowers RMSE by 0.246 log units, i.e. roughly 1.0×10^{-3} log units per molecule; going from 80 % to 100 % (an extra 163 molecules) lowers it by only 0.032, about 2.0×10^{-4} per molecule — a five-fold drop in marginal value. For a practitioner this quantifies *when data collection stops paying off* for a given model class, which is itself a scientifically useful output of the experiment rather than a mere accuracy curve.

4.4 E3 — Chemprop vs. Random Forest

Table 4 and Figure 6 compare the learned graph model against the descriptor-based Random Forest on identical splits.

Table 4: Chemprop MPNN vs. RDKit-descriptor Random Forest.

Dataset	Split	Model	RMSE	MAE	R^2
ESOL	random	Random Forest	0.802	0.549	0.864
ESOL	random	Chemprop MPNN	0.612	0.452	0.921
ESOL	scaffold	Random Forest	0.654	0.494	0.899
ESOL	scaffold	Chemprop MPNN	0.667	0.461	0.895
Lipophilicity	random	Random Forest	0.869	0.642	0.489
Lipophilicity	random	Chemprop MPNN	0.550	0.405	0.796
Lipophilicity	scaffold	Random Forest	0.853	0.649	0.499
Lipophilicity	scaffold	Chemprop MPNN	0.538	0.400	0.801

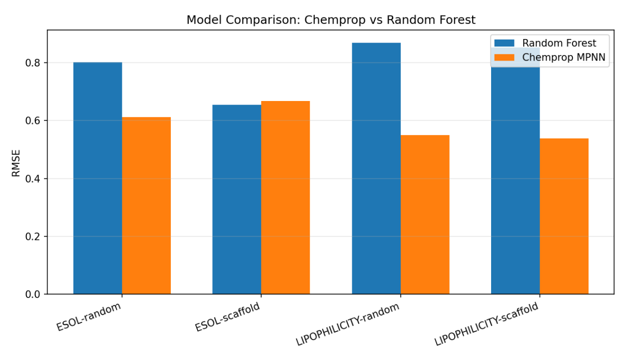

Figure 6: Chemprop MPNN vs. Random Forest RMSE across both datasets and both splits.

The learned representation provides a clear advantage under random splits: Chemprop reduces RMSE by 24 % on ESOL and by 37 % on Lipophilicity relative to the Random Forest. The gap is *largest on the larger dataset*: with 3,024 training molecules the MPNN has enough data to learn a representation that the eight hand-crafted descriptors cannot match (the Random Forest only reaches $R^2 \approx 0.49$ on Lipophilicity). On ESOL the gap is smaller and, under the scaffold split, the Random Forest and Chemprop are essentially tied (R^2 0.899 vs. 0.895). This is consistent with the descriptor-importance analysis: for ESOL the Random Forest places ≈ 0.81 of its importance on a single descriptor, Crippen $\log P$ — solubility is largely a lipophilicity-driven quantity, so a model given $\log P$ explicitly is already strong on a small dataset. For Lipophilicity the importance is spread across $\log P$, TPSA, molecular weight and FractionCSP3, and no descriptor subset suffices, which is exactly where the learned model wins.

4.5 E4 — Random split vs. scaffold split

The scaffold split forces the test scaffolds to be disjoint from the training scaffolds, a much harder and more realistic generalisation test. For **ESOL**, Chemprop RMSE rises from 0.612 (random) to 0.667 (scaffold) and R^2 falls from 0.921 to 0.895 — a modest, expected degradation: facing unseen skeletons is harder than interpolating among similar ones. For **Lipophilicity**, by contrast, Chemprop is essentially unchanged or marginally better under the scaffold split (RMSE 0.550 $\rightarrow$ 0.538, R^2 0.796 $\rightarrow$ 0.801). Two factors plausibly explain this property-dependent behaviour: (i) Lipophilicity is roughly four times larger, so even after enforcing scaffold disjointness the training set still covers a broad chemical space; and (ii) $\log D$ is governed more by bulk physicochemical character (size, polarity, lipophilic surface) than by the specific ring skeleton, so scaffold novelty hurts it less than it hurts a skeleton-sensitive property. The Random Forest shows the same qualitative pattern (ESOL even *improves* under the scaffold split here, R^2 0.864 $\rightarrow$ 0.899), confirming that the effect is a property/data characteristic rather than a model artefact.

The practical lesson is that average accuracy on a random split can *over-state* the reliability that a model will deliver on genuinely novel chemotypes, and that the size of this optimism is itself property-dependent. Scientific ML models should therefore be reported with a scaffold-split number, not only a random-split number. We also note a methodological subtlety: under the scaffold split the ESOL Random Forest *improves* relative to its own random-split score (R^2 0.864 $\rightarrow$ 0.899). This is not the model generalising better to novel chemotypes; rather, the particular scaffold partition happens to place a less dispersed set of solubilities in the test fold, so the R^2 denominator (total variance) changes. This illustrates why scaffold-split numbers should be read together with the predicted-vs- measured plot and the error distribution (§4.1–4.2), not as a single scalar — a recurring theme of this report.

4.6 Qualitative case study

As a sanity check we applied the trained ESOL and Lipophilicity models to ten well-known molecules (Figure 7 shows the ESOL subset). The predictions follow chemical intuition: highly hydroxylated, hydrophilic molecules such as glucose are predicted highly soluble ($\log S = +0.78$) and low-lipophilicity ($\log D = -2.20$), whereas hydrophobic steroids and aromatics such as testosterone ($\log S = -4.82$, $\log D = +3.57$) and tamoxifen ($\log S = -6.24$, $\log D = +3.95$) are predicted poorly soluble and highly lipophilic. The water/lipid contrast across the two models is internally consistent. We note that RDKit's $\log P$ and the model's $\log D$ are not identical concepts — $\log D$ depends on ionisation at a given pH — so disagreement for acidic/basic drugs (e.g. ibuprofen) is chemically reasonable rather than a model failure.

5. Discussion

Why more data helps, and why it saturates. The monotone improvement in §4.3 reflects the model progressively covering more of chemical space; the plateau between 80 % and 100 % indicates that, at this dataset size and model capacity, the marginal molecule carries little new structural information. The shrinking seed-variance with scale further shows that small molecular datasets produce not just less accurate but *less reproducible* models — an important caveat for small-data AI-for-Science claims. The per-molecule efficiency estimate in §4.3 (a five-fold drop in marginal value between the 20→50 % and 80→100 % regimes) turns this qualitative statement into an actionable one: it tells an experimentalist roughly where additional measurement effort stops translating into model accuracy for this property and model class.

Example ESOL Predictions

Molecule	Formula	Predicted LogS	Solubility
Aspirin	C9H8O4	-1.98	Moderate
Caffeine	C8H10N4O2	-0.64	Moderate
Benzene	C6H6	-1.20	Moderate
Ibuprofen	C13H18O2	-3.86	Low (poorly soluble)
Paracetamol	C8H9NO2	-1.13	Moderate
Glucose	C6H12O6	0.78	High (very soluble)

Figure 7: Example ESOL predictions for common molecules; predicted $\log S$ and the qualitative solubility label agree with chemical intuition.

What the error distribution tells us. The near-Gaussian, zero-centred ESOL residual (§4.2, $\sigma \approx 0.61$) is informative beyond its RMSE. Because its spread is on the order of the experimental reproducibility of solubility assays, the model is operating close to the *label noise floor*: the remaining error is increasingly irreducible by modelling alone and would instead require higher-quality or more consistent measurements. This reframes the usual "make the model bigger" reflex — for a mature benchmark such as ESOL, data quality, not model capacity, is the binding constraint. The mild negative-tail asymmetry (worst residual -2.27 vs. best $+1.46$) localises the residual risk to the sparsely sampled, very-insoluble extreme, exactly the region a practitioner should treat with caution.

Learned vs. engineered representations. Chemprop's advantage grows with dataset size and with the chemical complexity of the target. On small, $\log P$-dominated ESOL, a Random Forest given Crippen $\log P$ is already competitive; on larger Lipophilicity, where no single descriptor dominates, the learned representation is decisively better. The transparent baseline is therefore not redundant: it tells us *when* the expensive model is actually necessary, and its feature-importance profile (Crippen $\log P$ alone ≈ 0.81 on ESOL versus a flat profile on Lipophilicity) provides a cheap, interpretable diagnostic of *why* — a property whose variance is captured by one descriptor leaves little headroom for representation learning on a small dataset.

Chemical generalisation. The random-vs-scaffold contrast is the most research-relevant result. It demonstrates that a single accuracy number is insufficient to certify scientific reliability, and, more subtly, that the *cost of scaffold novelty is itself property-dependent*: skeleton-sensitive ESOL degrades while bulk-property-driven Lipophilicity does not. Reliability is bounded by data coverage *and* by the physics of the target property. The denominator artefact noted in §4.5 (the ESOL Random Forest's apparent scaffold-split *improvement*) reinforces a methodological point: R^2 is sensitive to the variance of whichever test fold a split produces, so robust evaluation should combine a scaffold split with the residual diagnostics of §4.1–4.2 rather than relying on any single scalar.

Practical recommendations. From these results we distil three concrete guidelines for small-to-medium molecular-property studies: (i) always report a transparent descriptor baseline alongside the graph model which both calibrates the gain and, through feature importance, explains it; (ii) report a scaffold-split number and an error distribution, not only a random-split RMSE, because the optimism of random splits is property-dependent and not knowable a priori; and (iii) use a data-scale curve to estimate marginal data value before committing to further data collection.

Limitations. All experiments use a single architecture and fixed hyper-parameters; we did not perform per-dataset tuning, ensembling, or uncertainty quantification, so the absolute numbers are conservative rather than state-of-the-art — the comparisons are controlled, not optimised. Both datasets are public benchmarks and may contain measurement noise and assay heterogeneity (particularly $\log D$, which is pH- and method-dependent). The scaffold split is a proxy for, not a substitute for, a genuine prospective external test set, and baseline runs in Tables 2 and 4 are single training runs, so we report point estimates without significance claims. These are natural directions for future work.

6. Conclusion

We studied a Chemprop message passing neural network across two ADMET properties, four training-data scales, two modelling paradigms and two splitting strategies. The MPNN predicts aqueous solubility ($R^2 = 0.921$) and lipophilicity ($R^2 = 0.796$) well, improves monotonically but with diminishing returns as data grows, beats a descriptor Random Forest under random splits (most decisively on the larger dataset), and shows a property-dependent — not catastrophic — response to scaffold splitting. The broader message for *AI for Science* is that data quantity, modelling paradigm and the chosen evaluation split jointly determine not just the accuracy but the *trustworthiness* of a scientific predictor. Future work should extend to more ADMET endpoints, model ensembling with calibrated uncertainty, and stricter external validation.

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